Exploratory Data Analysis (EDA) on Titanic Dataset

Step 1: Import Required Libraries

This step imports the required Python libraries for data manipulation, data visualization, and exploratory data analysis.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

[4] ✓ 3.5s Python

Exploratory Data Analysis (EDA)

Dataset Analysis Report

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Step 2: Load and Preview the Dataset

In this step, the Titanic dataset is loaded into a Pandas DataFrame using the `read_csv()` function. The `head()` function is then used to display the first five rows of the dataset, providing an initial overview of the data, column names, and sample records.

```
df = pd.read_csv("titanic.csv")
df.head()
```

[5] ✓ 0.1s Python

...

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

Step 3: Display Dataset Information

The `info()` function provides a summary of the dataset, including the total number of rows and columns, column names, data types, non-null values, and memory usage. This helps in understanding the dataset structure and identifying missing values.

```python
df.info()
```

[6] ✓ 0.0s Python

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    str    
 4   Sex          891 non-null    str    
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    str    
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    str    
 11  Embarked     889 non-null    str    
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB
```

Step 4: Generate Summary Statistics

The `describe()` function generates descriptive statistics for the numerical columns in the dataset. It provides information such as count, mean, standard deviation, minimum value, maximum value, and quartile values (25%, 50%, and 75%). This helps in understanding the distribution and spread of the data.

[7]

```python
df.describe()
```

✓ 0.0s

Python

...

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

Step 5: Check for Missing Values

The `isnull().sum()` function is used to identify and count missing values in each column of the dataset. Detecting missing values is an important step in Exploratory Data Analysis (EDA), as it helps determine whether data cleaning or preprocessing is required before further analysis.

```python
df.isnull().sum()
```

[8] ✓ 0.0s Python

```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

Step 7: Analyze Gender Distribution

The `value_counts()` function is used to count the number of passengers in each gender category. This helps understand the distribution of male and female passengers in the Titanic dataset and provides insights into the dataset's composition.

```
df["Sex"].value_counts()
```

[10] ✓ 0.0s Python

```
Sex
male      577
female    314
Name: count, dtype: int64
```

Step 6: Analyze Survival Distribution

The `value_counts()` function is used to count the number of passengers in each survival category. It shows how many passengers survived (`1`) and how many did not survive (`0`), providing an overview of the target variable's distribution.

```
df["Survived"].value_counts()
```

[9] ✓ 0.0s Python

```
Survived
0    549
1    342
Name: count, dtype: int64
```

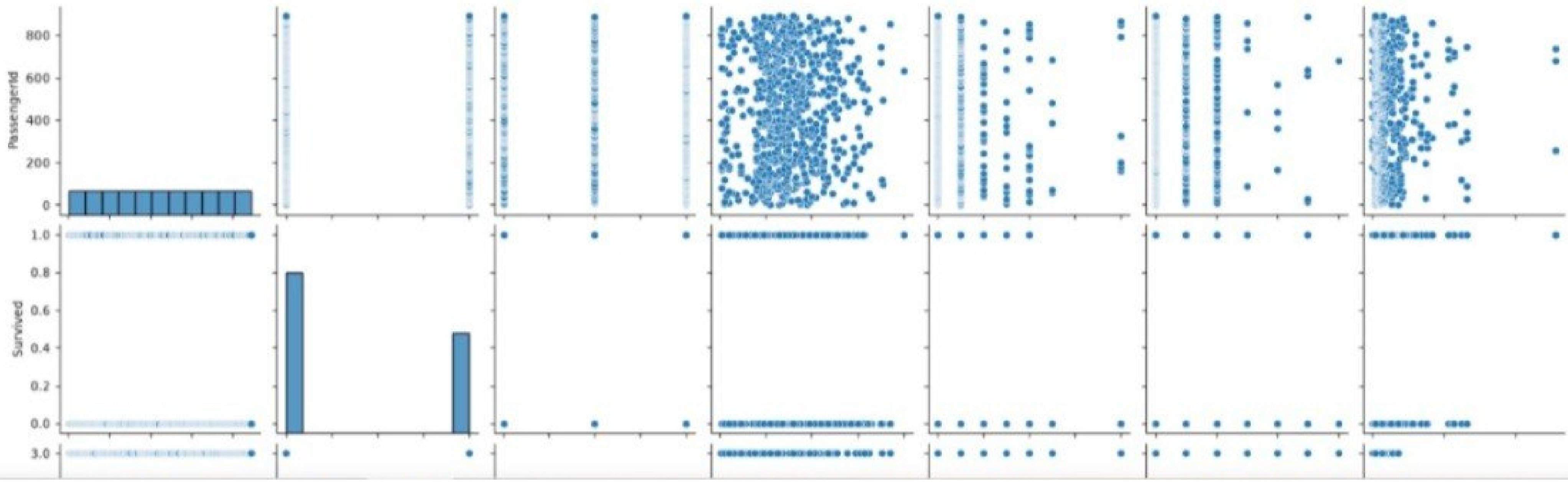
Step 8: Pair Plot

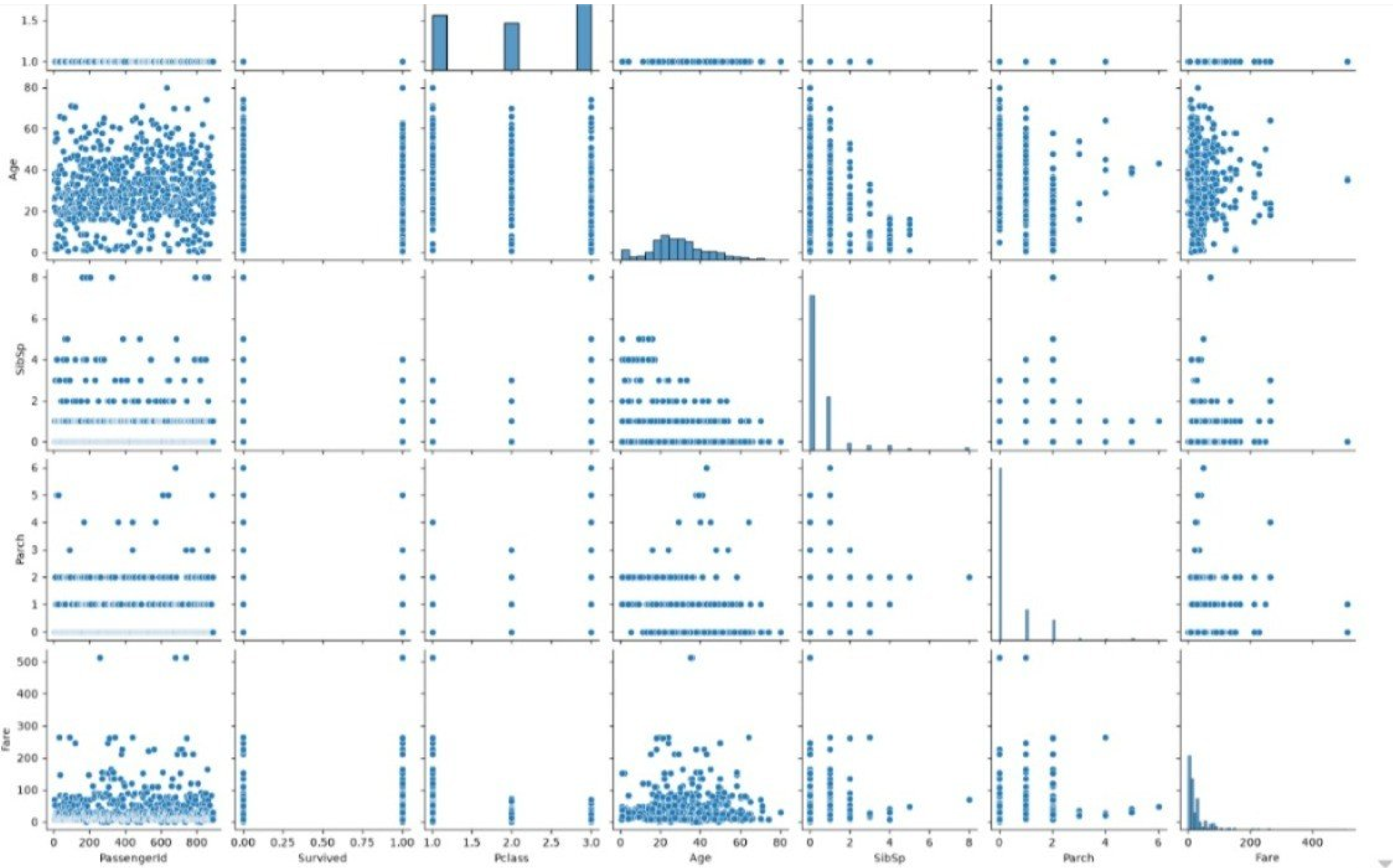
The pair plot visualizes the relationships between numerical variables and helps identify trends, patterns, and potential correlations.

Observation of PairPlot: The pair plot shows relationships among numerical features. Most variables do not show a strong linear relationship.

```
sns.pairplot(df)
plt.show()
```

[11] ✓ 7.9s Python





Age
Age
SibSp
Parch
Fare
PassengerId
Survived
Pclass
Age
SibSp
Parch
Fare

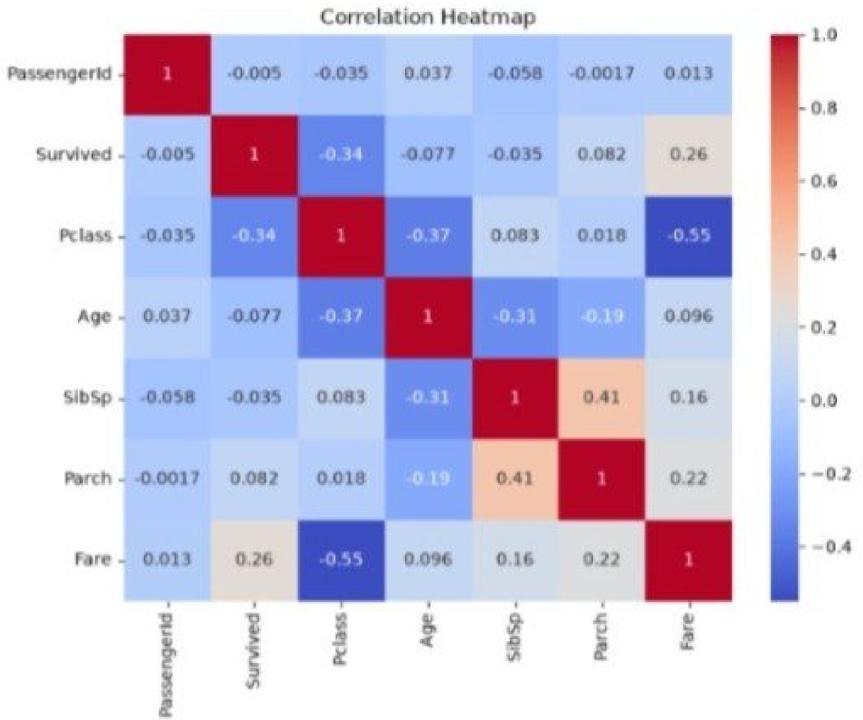
Step 9: Correlation Heatmap

The correlation heatmap shows the relationship between numerical features using correlation coefficients.

Observation of Heatmap: Most numerical features have weak correlations. Fare and Pclass show a noticeable negative correlation.

```python
plt.figure(figsize=(8,6))
sns.heatmap(df.select_dtypes(include='number').corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```

[12] ✓ 0.6s Python



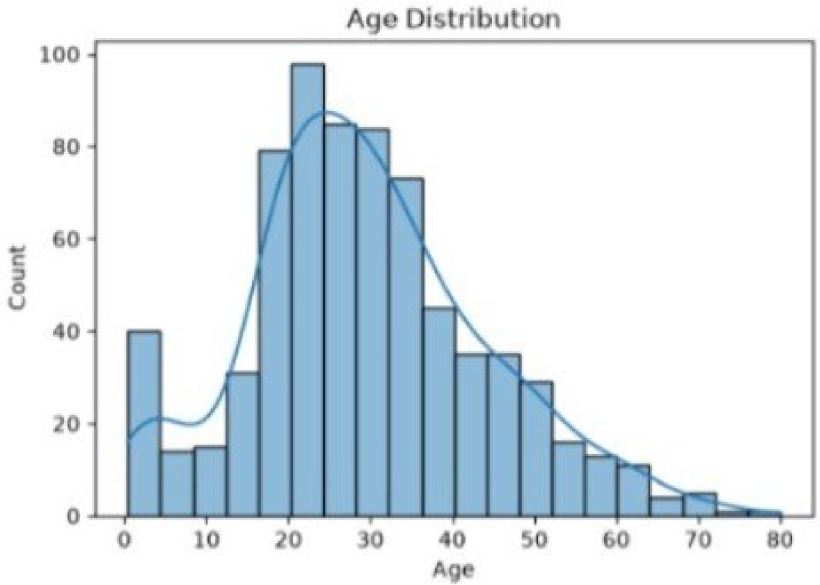
Step 10: Histogram of Age

A histogram is used to understand the distribution of passenger ages.

Observation of Histogram: Most passengers were between 20 and 40 years of age.

```python
plt.figure(figsize=(6,4))
sns.histplot(df["Age"], bins=20, kde=True)
plt.title("Age Distribution")
plt.show()
```

[13] ✓ 0.3s Python



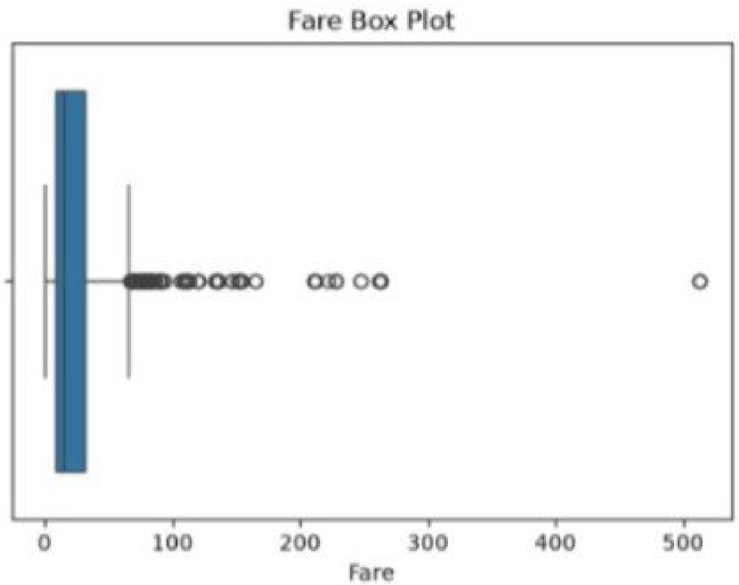
Step 11: Box Plot of Fare

A box plot is used to detect the spread of fare values and identify potential outliers.

Observation of Boxplot: The Fare column contains several outliers with very high ticket prices.

```
plt.figure(figsize=(6,4))
sns.boxplot(x=df["Fare"])
plt.title("Fare Box Plot")
plt.show()
```

[14] ✓ 0.2s Python



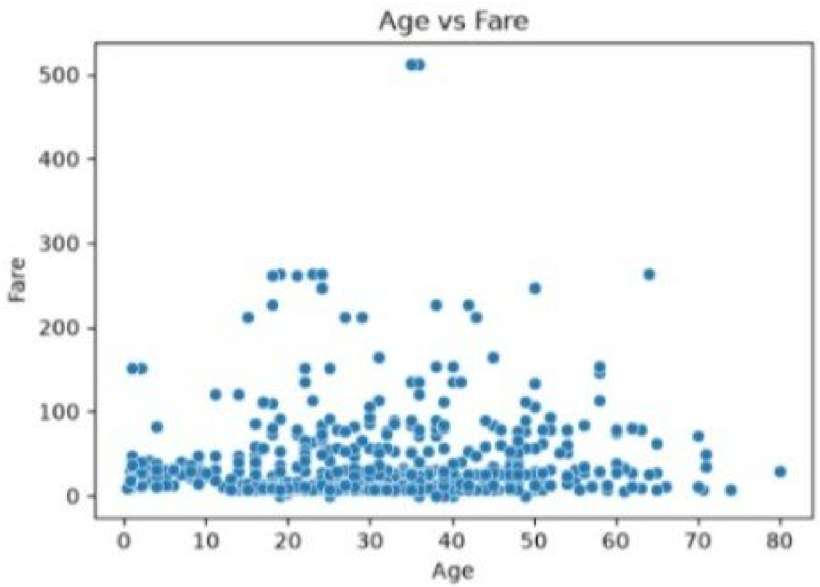
Step 12: Scatter Plot

The scatter plot shows the relationship between passenger age and fare.

Observation of ScatterPlot: There is no strong relationship between passenger age and ticket fare.

```
plt.figure(figsize=(6,4))
sns.scatterplot(x="Age", y="Fare", data=df)
plt.title("Age vs Fare")
plt.show()
```

[15] ✓ 0.2s Python



Summary of Findings

- The dataset contains missing values in columns such as Age, Cabin, and Embarked.
- Most passengers did not survive the disaster.
- Male passengers were more than female passengers.
- Fare values contain a few outliers.
- Age distribution is concentrated among young and middle-aged passengers.
- The heatmap indicates only weak to moderate correlations among numerical features.